SIMATS ENGINEERING
ASSESSMENT TOOL 2: Reverse Engineering

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: CSA1715

Assessment Tool Weightage: 30%

1. System Decomposition & Analysis

Analyze an existing AI-based system and **reverse engineer its architecture**, identifying key components, data flow, and underlying algorithms.

2. Architecture Reconstruction

Given a black-box software system, **reconstruct its system architecture** and explain the interaction between modules using appropriate diagrams.

3. Algorithm Identification

Reverse engineer a machine learning application to **identify the algorithms, data processing steps, and decision-making logic** used in the system.

4. Data Flow & Process Mapping

Examine an existing distributed AI application and **map its data flow, processing pipeline, and communication between components**.

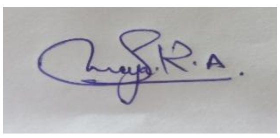
5. Improvement & Redesign

Perform reverse engineering on an intelligent system and **propose enhancements or redesign strategies** to improve scalability, efficiency, and performance.

Code of Conduct:

I Saranya.R.A *(192572052) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Evaluation Criteria	System Understanding	Decomposition & Analysis	Reconstruction of Architecture	Identification of Algorithms/Logic	Use of Tools & Techniques	Data Flow & Process Mapping	Problem-Solving & Interpretation	Innovation & Improvement Suggestions	Documentation & Presentation
Weightage	15%	15%	15%	10%	10%	10%	10%	5%	10%

Criteria	Excellent (5)	Good (4)	Average (3)	Poor (2)
System Understanding	Clearly identifies system purpose, components, and functionality	Mostly clear understanding	Basic understanding	Poor or incorrect
Decomposition & Analysis	Accurately breaks system into modules/components	Good decomposition	Partial breakdown	Incomplete analysis
Reconstruction of Architecture	Recreates system architecture with clear diagrams	Good reconstruction	Basic structure	Poor/incomplete
Identification of Algorithms/Logic	Clearly identifies underlying algorithms and logic	Mostly correct	Limited identification	Incorrect/missing
Use of Tools & Techniques	Effectively uses reverse engineering tools/techniques	Good usage	Limited use	No proper use
Data Flow & Process Mapping	Clearly explains data flow and interactions	Mostly clear	Some gaps	Poorly defined

Problem-Solving & Interpretation	Provides deep insights and correct interpretations	Good interpretation	Basic insights	Weak analysis
Innovation & Improvement Suggestions	Suggests meaningful improvements/redesign	Some suggestions	Basic ideas	No suggestions
Documentation & Presentation	Well-structured, clear diagrams and explanation	Mostly clear	Somewhat organized	Poor presentation

SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CO5 – ASSESSMENT TOOL 2

Reverse Engineering

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

QUESTIONS: 5 **Duration:** 1 Hour **Total Marks:** 50

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Understanding of Industry Problem	20% (2)	/2	/2	/2	/2	/2	/10
Application of Engineering Concepts	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/2.5	/12.5
Solution Design	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/2.5	/12.5
Use of Tools	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/1.5	/7.5
Analysis	10% (1)	/1	/1	/1	/1	/1	/5
Professionalism	5% (0.5)	/0.5	/0.5	/0.5	/0.5	/0.5	/2.5
Total per Question	**10 Marks**	**/10**	**/10**	**/10**	**/10**	**/10**	**/50**

100

ANSWER

1. System Decomposition & Analysis

a) System Identification

- Identify the purpose, objectives, and major functions of the existing AI-based system.
- Determine the type of users and applications supported by the system.

b) Component Identification

- Identify major components such as **user interface, application server, AI model, database, APIs, and storage**.
- Explain the purpose and responsibility of each component.

c) Input and Output Analysis

- Identify different types of input data provided to the system.
- Analyze the format, structure, and characteristics of the generated output.

d) Data Flow Analysis

- Trace the movement of data from input to preprocessing, AI processing, storage, and final output.
- Identify how data is transferred between different components.

e) Technology Analysis

- Identify possible programming languages, frameworks, databases, APIs, and cloud technologies used.

2. Architecture Reconstruction

a) Black-Box Analysis

- Study the external behavior of the system without directly accessing its source code.
- Analyze system inputs, outputs, responses, and functional behavior.

b) Module Identification

- Divide the system into logical modules such as **presentation, processing, AI/ML, database, and communication modules**.
- Determine the functionality of each module.

c) Module Interaction

- Analyze how different modules communicate with each other.
- Identify APIs, message queues, services, and data exchange mechanisms.

d) Architecture Diagram

- Construct a suitable **system architecture diagram** showing all major components.
- Use arrows to represent communication and data movement.

e) Architecture Validation

- Compare the reconstructed architecture with the observed system behavior.
- Verify whether the proposed architecture can explain the system's functionality.

3. Algorithm Identification

a) Data Preprocessing

- Identify data cleaning, missing-value handling, normalization, encoding, and transformation techniques.
- Determine how raw data is converted into usable input.

b) Feature Engineering

- Identify important features used by the AI system.
- Analyze feature selection and feature extraction techniques.

c) Algorithm Identification

- Determine the probable machine learning or deep learning algorithm.
- Consider algorithms such as **Decision Tree, Random Forest, SVM, Neural Network, CNN, RNN, or Transformer** based on system behavior.

d) Model Training

- Analyze how training data is provided to the model.
- Identify possible training, validation, and testing processes.

e) Prediction and Decision Logic

- Determine how the trained model generates predictions.
- Explain how predictions are converted into final decisions or recommendations.

4. Data Flow & Process Mapping

a) Data Collection

- Identify data sources such as **sensors, websites, mobile applications, databases, or APIs**.
- Analyze the frequency and volume of incoming data.

b) Data Ingestion

- Identify mechanisms used to collect and transfer data into the distributed system.
- Analyze possible technologies such as message queues and streaming platforms.

c) Data Storage

- Determine how structured, semi-structured, and unstructured data are stored.
- Identify possible distributed databases or cloud storage systems.

d) Data Processing

- Analyze batch and real-time processing activities.
- Identify distributed processing frameworks used for handling large datasets.

e) Communication and Data Flow

- Explain communication between different servers, services, databases, and AI models.
- Represent the complete process using a **Data Flow Diagram (DFD)** or pipeline diagram.

5. Improvement & Redesign

a) Performance Analysis

- Identify processing delays, memory limitations, network bottlenecks, and inefficient algorithms.
- Measure important factors such as response time and throughput.

b) Scalability Analysis

- Examine whether the system can handle increasing users and data volumes.
- Identify limitations in storage, computing resources, and network capacity.

c) Algorithm Improvement

- Replace inefficient algorithms with more suitable AI/ML techniques.
- Use model optimization and efficient feature-processing techniques.

d) Architecture Enhancement

- Introduce **cloud computing, distributed processing, load balancing, caching, and microservices** where appropriate.
- Improve communication between system components.

e) Security and Reliability

- Strengthen authentication, authorization, data protection, and access control.
- Add fault tolerance, backup, monitoring, and recovery mechanisms.

f) Proposed Redesign

- Develop an improved architecture based on the identified limitations.
- Compare the **existing and proposed systems** in terms of performance, scalability, cost, reliability, and efficiency.